

1.1 ABSTRACT

This project investigates the transport of temperature by means of convection and diffusion through one-dimensional domain using Computational fluid dynamics. The objective was to determine the non-transient temperature distribution over the domain by using the appropriate numerical scheme (solver) and comparing its accuracy with the analytical solution. We applied Finite differencing to discretize the convective and diffusive flux derivatives in the governing equation of the flow, to be more precise for the case of low cell Peclet number, diffusion transport was dominant across meshes so central differencing discretization scheme was applied because it gives symmetric weight to both the upstream and downstream values reflecting a diffusive characteristic. When the cell Peclet number was high, convection became dominant across a mesh therefore backward differencing was used because it reflects the directionality of information transport (from where the information came from) giving more weight to the upstream value. Dirichlet boundary conditions were defined as constants $T(x = 0) = T_0 = 0$ and $T(x = 1) = T_L = 1$, the uniform velocity over the domain was given for different mesh sizes: 0.1 m/s for 5 mesh elements, 2.5 m/s for 5 mesh elements, and 2.5 m/s for 20 mesh elements, the analytical solution is also given $\frac{T-T_0}{T_L-T_0} = \frac{e^{\rho u x / \Gamma} - 1}{e^{\rho u L / \Gamma} - 1}$, the domain length is 1 m, density of fluid ρ is uniform and set to 1 kg/m^3 , diffusive coefficient Γ is set to 0.1 kg/ms . After Convergence of the numerical solution using an iterative method and a good convergence criterion, it was compared to the analytical solution graphically and the following results were obtained: Finite Central differencing is not appropriate when cell Peclet number $Pe_{cell} > 2$ and produces non – physical oscillations (wiggles) and loss of boundedness. When mesh refinement was integrated to the problem, a great improvement was noticed using central differencing due to the decrease of Pe_{cell} from 5 to 1.2. Backward differencing provided better results than central differencing for the same Pe_{cell} of 5 which shows its effectiveness when dealing with convective prone problems. Furthermore, 2^{nd} order Backward differencing was tested against 1^{st} order and as expected it performed slightly better under the same conditions due to it being $2^{nd} \text{ order accurate}$.

1.2 PROBLEM DESCRIPTION

Temperature is being transported by convection and diffusion through a 1D domain without SOURCES, the governing equation of this flow is given by :

$$\rho U \frac{dT}{dx} = \Gamma \frac{d^2 T}{dx^2} \quad (1.1)$$

where ρ is the density, U is the velocity, T is the temperature, and Γ is the diffusion coefficient. The boundary conditions for this problem are $T(x = 0) = T_0 = 0$ and $T(x = 1) = T_L = 1$ We want to determine the temperature distribution over the domain for different velocity of flow using numerical schemes such as [Finite Differencing methods](#).

1.3 NUMERICAL METHOD

The governing equation (1.1) $\rho U \frac{dT}{dx} = \Gamma \frac{d^2 T}{dx^2}$ highlights two important derivatives, a 1st and 2nd order derivative for temperature, our goal is to convert those derivatives into algebraic equations so we can solve using an iterative method numerically for each of the cases stated below :

CASE	Velocity m/s	Number of Mesh
1	0.1	5
2	2.5	5
3	2.5	20

The derivatives are approximated using Taylor series expansion :

$$f(x + \Delta x) = f(x) + f'(x)\Delta x + \frac{f''(x)\Delta x^2}{2!} + \frac{f'''(x)\Delta x^3}{3!} + \dots \quad (1.2)$$

The 1st order derivative of Temperature can be approximated by either

- Central differencing :

$$T'(x) \approx \frac{T(x + \Delta x) - T(x - \Delta x)}{2\Delta x}, T'_i = \frac{T_{i+1} - T_{i-1}}{2\Delta x} \quad (1.3)$$

Or

- 1st order accurate Backward differencing which uses two points: x_i, x_{i-1} :

$$T'(x) \approx \frac{T(x) - T(x - \Delta x)}{\Delta x}, T'_i = \frac{T_i - T_{i-1}}{\Delta x} \quad (1.4)$$

- 2nd order accurate Backward differencing which uses three points: x_i, x_{i-1}, x_{i-2} :

$$T'(x) \approx \frac{3T(x) - 4T(x - \Delta x) + T(x - 2\Delta x)}{\Delta x}, T'_i \approx \frac{3T_i - 4T_{i-1} + T_{i-2}}{\Delta x} \quad (1.5)$$

The 2nd order derivative of Temperature can be approximated by :

- 2nd order accurate Central differencing which uses three points: x_i, x_{i-1}, x_{i+1}

$$T''(x) \approx \frac{T(x + \Delta x) - 2T(x) + T(x - \Delta x)}{\Delta x^2}, T''_i \approx \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta x^2} \quad (1.6)$$

The *order* of accuracy simply means that Δx is proportional to the *Error^{order}* of the numerical scheme meaning that a higher order of accuracy will produce more accurate results.

1.4 IMPLEMENTATION

To implement the discretizations scheme to the governing equation (EQN(1.1)) we just substitute one from those stated above. To choose the appropriate discretization we have to look at a very important parameter of this simulation, which is the **Cell Peclet number**. This number tells which physics is dominant across a Cell, DIFFUSION or CONVECTION, so based on this information we can choose between Central differencing which treats diffusion problems better or Backward differencing which treats convection better.

Depending on the value of the Péclet number, Pe :

- $Pe \ll 1$: diffusion dominates
- $Pe \approx 1$: convection and diffusion are comparable
- $Pe \gg 1$: convection dominates

The governing equation is

$$\rho U T'_i = \Gamma T''_i. \quad (1.7)$$

Notice, that we have two terms, on the left is the convective term and on the right is the diffusive term, for the diffusive term we will always use Central differencing because we know that it reflects a diffusive behavior giving equal weights to the upstream and downstream points, on the other side we can use central differencing on the convective term under appropriate conditions.

Finding this condition is actually finding the constraint on the Cell Peclet number such that central differencing does not produce non-physical oscillations because they will be amplified resulting in an overflow (inf. values) in the solution due to the iterative nature of the solving process.

Substitute EQUATION(1.3) and EQUATION(1.6) into EQUATION(1.7) and make T_i subject of formula gives:

$$T_i = T_{i+1} \left(\frac{1}{2} - \frac{Pe_{cell}}{4} \right) + T_{i-1} \left(\frac{1}{2} + \frac{Pe_{cell}}{4} \right) \quad (1.8)$$

For BOUNDEDNESS of the numerical solution we require **NON-Negative** coefficients because negative coefficients can lead to overshoots and undershoots of the solution and in our case it's more dangerous because we are using iterative methods, so if it's negative it might get more negative and so on to infinity.

In this flow $u > 0$ and $Pe_{cell} = \frac{\rho u \Delta x}{\Gamma}$ which implies $Pe_{cell} > 0$

Therefore we require $(\frac{1}{2} - \frac{Pe_{cell}}{4}) > 0$ AND $(\frac{1}{2} + \frac{Pe_{cell}}{4}) > 0$ for the solution to be bounded which can be rewritten as $-2 < |Pe_{cell}| < 2$.

Hence the Cell Peclet number should be $-2 < |Pe_{cell}| < 2$ for FINITE CENTRAL DIFFERENCING to work for the convective term otherwise we will have to use FINITE BACKWARD DIFFERENCING as it accounts for direction of information transport , i.e it gives more weight to the upstream value.

Substitute EQUATION(1.4) and EQUATION(1.6) into EQUATION(1.7) and make T_i subject of formula gives:

$$T_i = T_{i+1}(\frac{1}{2 + Pe_{cell}}) + T_{i-1}(\frac{1 + Pe_{cell}}{2 + Pe_{cell}}) \quad (1.9)$$

This is the equation we get when we use 1st order accurate Backward differencing on the convective term and 2nd order accurate Central differencing on the diffusive term. A better version of this equation can be obtained if we use 2nd order accurate Backward differencing for the convective term.

Substitute EQUATION(1.5) and EQUATION(1.6) into EQUATION(1.7) and make T_i subject of formula gives:

$$T_i = \frac{2T_{i+1} + (2 + 4Pe_{cell})T_{i-1} - Pe_{cell}T_{i-2}}{3Pe_{cell} + 4} \quad (1.10)$$

This equation will give better results than (1.9) because its 2nd order accurate.

NOTE that $T_i = T[i]$ numerically.

To implement those equation into the simulation, we first need to convert the 1D domain into a 1D computational mesh (grid of points) and apply the discretized equation eqn.(1.8),eqn.(1.9),eqn.(1.10) at each of the mesh grid points. We evaluate the temperature at those points iteratively until it stops changing locally which in other words means a **CONVERGENCE** has been reached. For the convergence criterion we use the sum of the absolute errors between our previous Value T and the new iterated T_{new} for all the points and if its less than $\epsilon = 10^{-8}$ the numerical solution has converged.

1.5 VERIFICATION

Comparing Numerical results to analytical

The following data holds true for the 3 cases: $L = 1.0m, \rho = 1.0kg/m^3, \Gamma = 0.1kg/m.s$

1. **Table 1.1** represents the data gathered after the simulation of **CASE 1**-refer to the table in *section 1.3 NUMERICAL METHOD*: $U = 0.1m/s$ and 5 Mesh. For **CASE 1** the domain peclet number is $Pe_L = \frac{\rho u L}{\Gamma} = 1$ which means convection and diffusion strength are equally comparable over the domain. The cell peclet number is $Pe_{cell} = \frac{\rho u \Delta x}{\Gamma} = 0.2$ which is $\ll 1$ therefore diffusion transport dominates across a cell(mesh) and also the $-2 < |Pe_{cell}| < 2$ which says that central differencing can be used for the convective term. EQUATION (1.8) is used for CASE 1.

Table 1.1: Numerical and analytical solutions along the domain for case 1.

Node	Distance	Numerical solution	Analytical solution	Absolute error
1	0.0	0.0	0.0	0.0
2	0.2	0.128645	0.128857	0.000213
3	0.4	0.285877	0.286241	0.000364
4	0.6	0.478049	0.478467	0.000418
5	0.8	0.712927	0.713247	0.00032
6	1.0	1.0	1.0	0.0

Node identifies the grid point, Distance is the x -value on the one-dimensional domain, Numerical solution is the value obtained from the iterative numerical method, Analytical solution is the exact solution on the domain of length 1 m evaluated at each x -value, and Absolute error is the difference between the two solutions.

2. **Table 1.2** represents the data gathered after the simulation of **CASE 2** $U = 2.5m/s$ and 5. For **CASE 2** the domain peclet number is $Pe_L = \frac{\rho u L}{\Gamma} = 25$ which means convection transport is way stronger than diffusion transport over the domain. The cell peclet number is $Pe_{cell} = \frac{\rho u \Delta x}{\Gamma} = 5$ which is $\gg 1$ therefore convection transport dominates across a cell(mesh) and also since $Pe_{cell} > 2$ shows that central differencing will produce non-physical oscillations if used for the convective term. EQUATION (1.9)-1st order accurate Backward differencing is therefore used for CASE 2.

Table 1.2: Numerical and analytical solutions along the domain for case 2.

Node	Distance	Numerical solution 01	Analytical solution	Absolute error
1	0.0	0.0	0.0	0.0
2	0.2	0.000643	0.0	0.000643
3	0.4	0.004502	0.0	0.004501
4	0.6	0.027653	4.6e-05	0.027607
5	0.8	0.166559	0.006774	0.159785
6	1.0	1.0	1.0	0.0

3. **Table 1.3** represents the data gathered after the simulation of **CASE 2** using 2^{nd} order accurate Backward differencing for the convective term : $U = 2.5m/s$ and 5 Mesh.

Table 1.3: Numerical and analytical solutions along the domain for case 2.

Node	Distance	Numerical solution 01	Analytical solution	Absolute error
1	0.0	0.0	0.0	0.0
2	0.2	0.000643	0.0	0.000643
3	0.4	0.004502	0.0	0.004501
4	0.6	0.027653	4.6e-05	0.027607
5	0.8	0.166559	0.006774	0.159785
6	1.0	1.0	1.0	0.0

The 2^{nd} order accurate Backward differencing for the convective term acutally produces slightly better results than the 1^{st} order as when we compare the total Absolute error for all the nodes of Table 2 with Table 3 we see that $total_{error}$ case 2- 2^{nd} $ORDER = 0.132401$ against $total_{error}$ case 2- 1^{st} $ORDER=0.192536$, clearly $0.132401 \ll 0.192536$.

4. **Table 1.4** represents the data gathered after the simulation of **CASE 3** $U = 2.5m/s$ and 20. For **CASE 3** the domain peclet number is $Pe_L = \frac{\rho u L}{\Gamma} = 25$ which means convection transport is way stronger than diffusion transport over the domain. The cell peclet number is $Pe_{cell} = \frac{\rho u \Delta x}{\Gamma} = 1.25$ which is $\gg 1$ therefore convection transport slightly dominates across a cell(mesh) and also since $Pe_{cell} < 2$ shows that central differencing is appropriate for the convective term. EQUATION (1.8)- FINITE CENTRAL differencing is therefore used for CASE 3.

The results obtained are very close to the exact solution.

Table 1.4: Numerical and analytical solutions along the domain for case 3.

Node	Distance	Numerical solution	Analytical solution	Absolute error
1	0.0	0.0	0.0	0.0
2	0.05	0.0	0.0	0.0
3	0.1	0.0	0.0	0.0
4	0.15	0.0	0.0	0.0
5	0.2	0.0	0.0	0.0
6	0.25	0.0	0.0	0.0
7	0.3	0.0	0.0	0.0
8	0.35	0.0	0.0	0.0
9	0.4	0.0	0.0	0.0
10	0.45	0.0	1e-06	1e-06
11	0.5	0.0	4e-06	3e-06
12	0.55	2e-06	1.3e-05	1.1e-05
13	0.6	8e-06	4.6e-05	3.8e-05
14	0.65	3.5e-05	0.00016	0.000125
15	0.7	0.000151	0.000557	0.000406
16	0.75	0.000654	0.001943	0.001288
17	0.8	0.002836	0.006774	0.003938
18	0.85	0.012289	0.023619	0.01133
19	0.9	0.053254	0.082337	0.029083
20	0.95	0.230769	0.286974	0.056204
21	1.0	1.0	1.0	0.0

1.6 Results and discussion

Three cases were analysed to investigate Temperature distribution over a domain. The cases differ in flow speed and mesh number. The results for each case are presented below, followed by a comparison of the observed trends.

1. Case 1 : Figure 1 shows the variation of temperature with position. The results indicate weak exponential decay from the main temperature source $T=1$ at $x = L = 1$. Temperature spreads over the whole domain ,all the interior points have a temperature > 0 because according to the Domain peclet number convection and diffusion transport has the same strength therefore temperature gets carried away stronger from the main source $T=1$ by convection and diffusion. The maximum value of Temperature occurs at $x = L$. This behaviour can be attributed to convection and diffusion being equally compared. The Cell peclet number is $-2 < |Pe_{cell}| < 2$ which indicates that central differencing will work for the convective term and we will end up having a bounded solution.

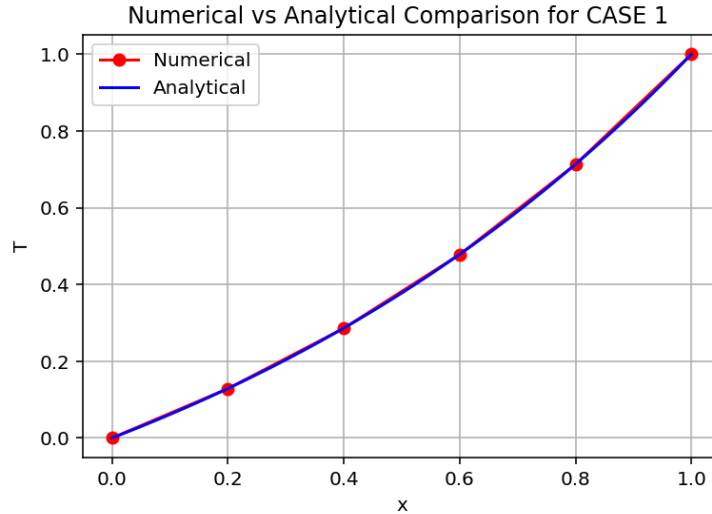


Figure 1.1: Numerical and analytical solutions for case 1.

2. Case 2: The results indicate strong exponential decay from the constant Temperature source at $x = L$, this is expected because the Domain peclet number $Pe_L \gg 1 = 25$. This high peclet number indicates that convection is dominant on the whole domain . If $\frac{u}{\Gamma}$ is written as a constant c then the following differential eqn $T'' = cT'$ has the general solution $T = C_1 + C_2 \exp^{ax}$ therefore we can rewrite this as $T = C_1 + C_2 \exp^{\frac{Pe_L x}{L}}$. So , if $Pe_L \gg 1$ then the exponential will change very fast over the domain , this is why for high peclet number when convection is dominant we see this

Strong exponential growth.

Numerical 1st order and 2nd order vs Analytical Comparison for case 2 using 1st order backward scheme

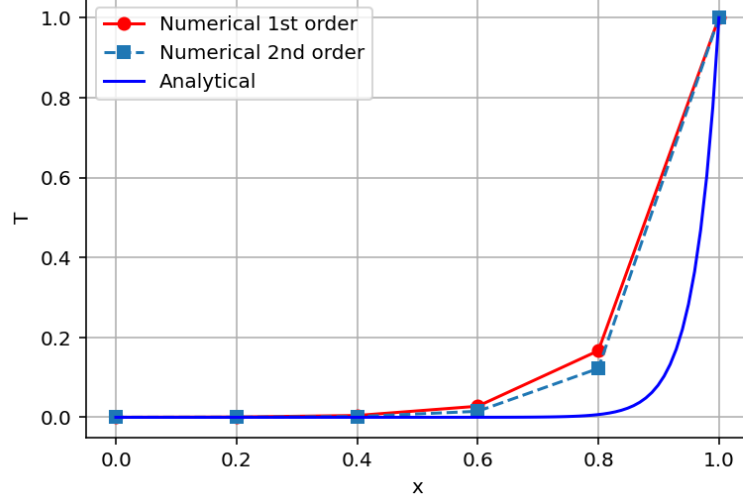


Figure 1.2: Numerical and analytical solutions for case 2.

Even better results can be obtain using a 2^{nd} order accurate Backward differencing for the convective term

- Case 3: For this case the Domain peclet number does not change it stays the same as case 2 i.e $Pe_L = 25$ but mesh reffinement is performed which changes the mesh number from 5 to 20 decreasing the grid spacing Δx from 0.2 to 0.05 , this acutally favours Central difference scheme because the Cell Peclet number Pe_{cell} drops from 5 to 1.25 and therefore $-2 < |Pe_{cell}| < 2$. This means that we should obtain a better numerical approximation than case 2 because the lower the grid spacing Δx the more accurate the simulation.

Numerical vs Analytical Comparison for case 3 using Central differencing scheme after mesh refinement

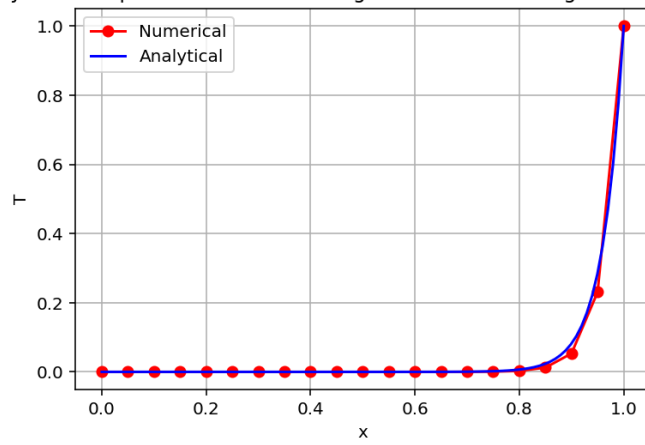


Figure 1.3: Numerical and analytical solutions for case 2.

1.7 Conclusion

The simulations of the 1D convection-diffusion equation using Finite Central differencing scheme was a very good approximation of the Analytical solution but the iterative method used can amplify non-physical oscillations and cause loss of boundedness in some cases where the inappropriate discretization scheme is used, for example using Finite central difference for the convective term when the $Pe_{cell} > 2$ will result in inf. in the solution array which indicates overflow, this is why we should always verify the Péclet number to identify the physics which dominates the flow and then choose the correct Discretisation scheme.